PLSC 504 - Topics in Political Methodology

Exercise Seven

November 3, 2026

Introduction

This exercise is focused on item response theory. As intimated in class, it has no simulation component.

Exercise

For nearly 100 years, researchers have used the votes of the justices to measure their ideological positions. The current state of the art in this area are the Martin and Quinn (2001) scores (`https://mqscores.lsa.umich.edu/`). Those scores are conceived and estimated in a Bayesian framework, and use an autoregressive term to allow a justice's scores to change over time. Here, we're going to consider a couple alternative ideas, using an IRT approach.

We'll be working with a "wide" version of the Supreme Court Database (`https://scdb.la.psu.edu/`). These are data on every U.S. Supreme Court decision (and the votes of the justices therein) for the 1946-2025 Terms;[1] each line of data is a single case (docket number), while columns record information about that case. The codebook with details about the data is available at that website. In addition to some case-level variables (docket number, U.S. citation, case name, term, decision date, sitting Chief Justice, issue area, the direction of the overall decision, the overall case disposition, and the number of majority and minority votes), the data have three sets of variables that are central to this exercise:

- `ZZZDirection` variables, indicating the the *ideological direction* of each justice's vote (0 for a "liberal" vote, 1 for a "conservative" vote).
- `ZZZMajority` variables, indicating whether (=1) or not (=0) a justice's vote was with the *majority* in the decision.
- `ZZZReverse` variables, indicating whether (=1) or not (=0) the justice's vote was to *reverse* the lower court's decision.

For all three of these variable types, "ZZZ" denotes the name of the justice in question, with first-initial identifiers to remove ambiguity. So, for example, the variable indicating the ideological direction of Chief Justice Roberts' vote is called `RobertsDirection`.

[1] SCOTUS terms run from October through June or July.

Your assignment is as follows:

1. First, fit a standard two-parameter logistic IRT model to generate estimates of the ideological positions of the justices. Do so using the `Direction` variables. Present and briefly discuss your results, including a discussion of any potential concerns.

2. Leaving ideology aside for the moment, consider the information contained in the `Majority` variables. Suppose for a moment that this reflects the extent of "agreeableness" (or, perhaps, "conformity") of a justice, in that a justice who appears in the majority often is more agreeable / conformist than one who is often in the minority. Fit one or more IRT model(s) of the degree of agreeableness/conformity, and briefly present and discuss those results. Also, compare the `Direction` and `Majority` results, and speculate on what, if anything, might be going on between the two.

3. In a 2009 paper, McGuire et al.[2] note the presence of "affirmance bias" on the Supreme Court: a situation where the association between justices' ideologies and their votes is significantly stronger in cases where the Court reverses the lower court's decision than in cases resulting in an affirmance. In this part, fit and compare separate measurement models of the justices' ideology for cases where the justice voted to affirm the lower court decision and those where s/he voted to reverse it. Discuss the differences you find, both statistically and substantively, and speculate a bit on the bases for those differences.

This homework is due *electronically* (in the usual fashion: emailed to Ludmilla and me) by 11:59 p.m. ET on **Tuesday, November 17, 2026**, and is worth the usual 50 points.

[2]McGuire, Kevin T., Georg Vanberg, Charles E. Smith, and Gregory A. Caldeira, "Measuring Policy Content on the U.S. Supreme Court." *Journal of Politics* 71:1305-1321.